

Willian Vieira PhD

Statistical Modeler and Data Engineer with a PhD in Ecology. I build machine learning pipelines on large-scale geospatial and remote-sensing data — from probabilistic models to the production infrastructure that runs them.

DATA SCIENCE & ENGINEER EXPERIENCE

2025
|
2024
(1 yr 9 m)

• Data Analyst

Habitat, Montreal, Canada

Architected end-to-end R/Python ETL pipelines to deploy geospatial ML models into production environments | Built and maintained reproducible pipelines processing large-scale satellite imagery and multi-source spatial data for continental monitoring | Designed a geospatial lakehouse architecture for streaming unstructured datasets from Azure cloud storage, optimizing data retrieval for analytics and ML training | Implemented metadata-driven workflows and automated validation for training-dataset governance and pipeline reliability | Led the transition to a Unix-based production environment — Docker, CI/CD, automated setup scripts, DevOps best practices

2024
|
2017

• PhD Research

Integrative Ecology Lab, Sherbrooke, Canada

Built end-to-end geospatial data pipelines to ingest, clean, and harmonize continental-scale forest-climate datasets for Bayesian hierarchical modelling | Developed reproducible HPC (High-Performance Computing) workflows (, , ) leveraging parallel processing and cluster management to execute thousands of model simulations at scale | Developed open-source software libraries (, ) for geospatial demographic modelling, implementing version control and automated testing | Published a **technical methods book** documenting the full computational pipeline to ensure long-term project reproducibility, along with one peer-reviewed **publication** and two preprints (, )

2022
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2020
(part-time)

• Biostatistician

Environment and Climate Change Canada - Quebec, Canada

Developed a cost-aware probability sampling protocol to optimize the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias correction methodologies, designing a method later adopted by other provinces | Engineered a fully automated and reproducible **geospatial pipeline** featuring automated documentation and version-controlled workflows to ensure long-term system maintainability | Produced a fully open source, reproducible **analytical workflow** with automated documentation

EDUCATION

2024
|
2017

• PhD, Ecology

Université de Sherbrooke - Sherbrooke, Canada

 *How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations*

2016
|
2015

• Masters 2, Agroecology and Resource Management

Bordeaux Sciences Agro, Bordeaux, France

 *Modelling the dispersion of weed species in agricultural landscapes*

2015
|
2010

• BSc in Agronomy

Universidade Federal de Santa Catarina, Florianópolis, Brazil

CONTACT

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 [LinkedIn/WillVieira](#)

 [GitHub/WillVieira](#)

 [Publications](#)

SKILLS

Statistics & Modelling:

Machine learning · Bayesian inference · Hierarchical models · Probabilistic modeling · Spatial statistics · Causal inference · Experimental design

Programming & Database:

R · Python · Stan · Julia · DuckDB · PostGIS · Bash · SQL

Data Engineering & Cloud:

ETL/ELT · Azure · AWS · Docker · Apache Arrow · STAC · COG · GeoParquet · HPC · Remote sensing

DevOps & Workflow:

Git · CI/CD · Unix · Unit testing · Make · Quarto · Jupyter

OUTREACH

· Developed and maintained the **QCBS R Workshop Series**, reaching nearly one thousand graduate students · Taught +250 students across biology, engineering, and graduate programs (programming, statistics, ecology) · **5 published** papers & preprints with ~100 citations · Translated client needs into technical analyses and delivered clear, actionable results

LANGUAGES

Portuguese · Native
French · Full Professional
English · Full Professional

CV source code hosted on 

Last updated: 2026-05-06

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